Introduction to Computer Control Systems, 5 credits, 1RT485

Date: 2025-06-11

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Number of problems: 5

Allowed aid: A non-programmable calculator and mathematical handbooks

Preliminary grades:	grade 3	15 points
	grade 4	21 points
	grade 5	26 points

Some general instructions and information:

- Your solutions should be given in English.
- Write only on one side of the paper.
- Write your exam code and page number on all pages.
- Do not use a red pen.
- Use separate sheets of paper for the different problems (i.e. the numbered problems, 1–5).

With the exception of Problem 1, ***all your answers must be clearly motivated!*** *A correct but unmotivated answer will score reduced points!*

Best of luck!

Formula sheet

Laplace transform table

Table 1: Basic Laplace transforms

$f(t)$	$F(s)$	$f(t)$	$F(s)$
unit impulse — imp(t)	1	$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$
unit step — step(t)	$\frac{1}{s}$	$e^{-at}\cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$
unit ramp — ramp(t)	$\frac{1}{s^2}$	$\sinh(bt)$	$\frac{b}{s^2-b^2}$
t	$\frac{1}{s^2}$	$\cosh(bt)$	$\frac{s}{s^2-b^2}$
t^n	$\frac{n!}{s^{n+1}}$	$\frac{1}{2b}t\sin(bt)$	$\frac{s}{(s^2+b^2)^2}$
e^{-at}	$\frac{1}{s+a}$	$t\cos(bt)$	$\frac{s^2-b^2}{(s^2+b^2)^2}$
$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$	$\frac{\cos(bt)-\cos(at)}{a^2-b^2}; (a^2 \neq b^2)$	$\frac{s}{(s^2+a^2)(s^2+b^2)}$
$\frac{1}{(n-1)!}t^{n-1}e^{-at}; (n=1,2,3...)$	$\frac{1}{(s+a)^n}$	$\frac{\sin(at)+at\cos(at)}{2a}$	$\frac{s^2}{(s^2+a^2)^2}$
$\sin(bt)$	$\frac{b}{s^2+b^2}$		
$\cos(bt)$	$\frac{s}{s^2+b^2}$		

Table 2: Properties of Laplace Transforms

$\mathcal{L}[af(t)] = aF(s)$	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
$\mathcal{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$	$\mathcal{L}\left[t^2 f(t)\right] = \frac{d^2}{ds^2}F(s)$
$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s)-f(0)$	$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}F(s), \quad n \in \mathbb{N}$
$\mathcal{L}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s)-sf(0)-f'(0)$	$\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$
$\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}+\frac{1}{s}\left[\int f(t)\,dt\right]_{t=0}$	$\mathcal{L}\left[\int_0^t f_1(t-\tau)f(t)d\tau\right] = F_1(s)F_2(s)$
$\mathcal{L}[f(t-a)] = e^{-as}F(s)$	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$

Open-loop and sensitivity functions

$$G_o(s) = G(s)F_y(s), \qquad S(s) = \frac{1}{1+G_o(s)}, \qquad T(s) = 1-S(s) = \frac{G_o(s)}{1+G_o(s)}$$

State-space forms and transfer function relations

- State-space form and transfer function

$$\begin{aligned} \dot{x} &= Ax+Bu \\ y &= Cx+Du \end{aligned} \quad \Rightarrow \quad \boxed{G(s) = C(sI-A)^{-1}B+D}$$

- Associated matrices

$$\mathcal{S} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- LTI system with transfer function

$$\boxed{G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}}$$

  i) Observable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

  ii) Controllable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} b_1 - a_1 b_0 & b_2 - a_2 b_0 & \cdots & b_n - a_n b_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

- Solution to state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0$$

  can be written as

$$\boxed{x(t) = e^{At} x_0 + \int_0^t e^{A\tau} B u(t-\tau) d\tau}$$

- Observer system

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})}$$

Feedback control structures

General linear feedback in Laplace form:

$$\boxed{U(s) = F_r(s)R(s) - F_y(s)Y(s)}$$

Common control structures in this form.

- PID controller:

$$F_y(s) = F_r(s) = F(s) = K_p + \frac{K_i}{s} + K_d s,$$

  where $K_p, K_i, K_d \geq 0$

- State-feedback controller with observer:

$$\begin{aligned} F_r(s) &= \left(1 - L(sI - A + KC + BL)^{-1}B\right)\ell_0 \\ F_y(s) &= L(sI - A + KC + BL)^{-1}K \end{aligned}$$

Bode's phase diagram

Consider the loop gain $G_o(s) = \mu \frac{\prod_{k=1}^{n_z}(1-\tau_k s)}{s^g \prod_{k=1}^{n_p}(1-T_k s)}$, where for simplicity $\mu > 0$. The phase of $G(i\omega)$ at a given frequency is

$$\arg\left(G_o(i\omega)\right) = -g\,90^\circ + \left[\sum_{k=1}^{n_z} \arctan\left(\tau_k \omega\right) - \sum_{k=1}^{n_p} \arctan\left(T_k \omega\right)\right] \frac{180^\circ}{\pi}$$

Routh–Hurwitz stability criterion

Let $\Phi(s) = a_0 s^n + b_0 s^{n-1} + a_1 s^{n-2} + ...$ be the characteristic polynomial.

$$\begin{array}{ccc} a_0 & a_1 & ... \\ b_0 & b_1 & ... \\ \hline c_0 & c_1 & ... \\ \vdots & \ddots & \end{array}$$

A generic element of the third row, c_k, can be computed as

$$c_k = \frac{b_0 a_{k+1} - a_0 b_{k+1}}{b_0}$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$\boxed{\begin{aligned} x_k &= Fx_k + Gu_k \\ y_k &= Hx_k \end{aligned}}$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$
\boxed{\begin{aligned} x(k+1) &= Fx(k) + Gu(k) \\ y(k) &= Hx(k) \end{aligned}}
$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Bilinear transform formula ($\alpha \in \{0, \frac{1}{2}, 1\}$)

$$
s = \frac{1}{T} \frac{1 - q^{-1}}{\alpha + (1 - \alpha)q^{-1}}
$$

Problem 1: basic questions (6/30)

Answer only *'true'* or *'false'*. Each correct answer gives 1 point, each wrong answer gives -1 point. Minimum total points for Part A, B and C is 0.

Part A

Note: Write 'skip' if your home assignment 1 score ≥ 6.5

i) If a linear system is stable, a sinusoidal input yields an (asymptotically) sinusoidal output at the same frequency.

ii) The root locus shows how the poles of the open-loop transfer function $G_o(s) = KG(s)$ evolve as K increases.

iii) If $G_o(0) = 1$, the output tracking error is asymptotically null.

(3 p)

Part B

Note: Write 'skip' if your home assignment 2 score ≥ 6.5

i) Given a transfer function $G(s)$, the controllable and observable canonical forms are the only minimal state-space realizations.

Consider now the closed-loop system shown in Figure 1

ii) A measurement noise $n(t) = \sin(40t)$ is attenuated by a factor 100.

iii) A disturbance $v(t) = \sin(0.1t)$ gets passed (i.e., its attenuation factor is 1).

(2 p)

Part C

Note: Write 'skip' if you attended/completed the fourth computer session (D4)

i) When discretizing a controller, a lower sampling time always leads to a smaller loss of phase margin.

(1 p)

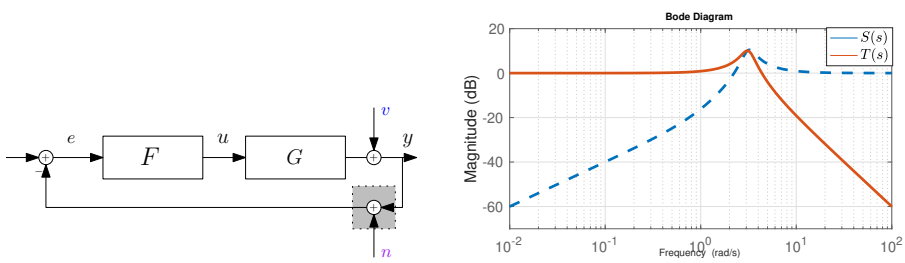

Figure 1: On the left, the closed-loop system affected by the measurement noise $n(t)$ and by the disturbance v. On the right, the bode plot of its sensitivity $S(s)$ (dashed blue line) and complementary sensitivty $T(s)$ functions (solid red line).

Proposed solution to problem 1

Part A

i) True

ii) False

iii) False

Part B

i) False

ii) True

iii) False

Part C

i) True

Problem 2 (6/30)

Consider the system described by the following transfer function

$$G(s) = \frac{s - 0.25}{s^2 + 2s + 1}$$

Consider the proportional controller $F(s) = K$.

a) For what values of $K \in \mathbb{R}$ does the controller yield a closed-loop stable system?

(3p)

b) Consider a step reference signal $r(t) = R \cdot \text{step}(t)$. For what values of K is the steady-state output error smaller than 10%?

(3p)

Proposed solution to problem 2

a) The closed-loop transfer function reads

$$G_c(s) = \frac{KG(s)}{1 + KG(s)} = \frac{K(s - 0.25)}{s^2 + (2 + K)s + 1 - 0.25K}$$

This transfer function is stable as long as the roots of the denominator have negative real part. This is the case when

$$\begin{cases} 2 + K > 0 \\ 1 - 0.25K > 0 \end{cases}$$

The solution is therefore $-2 < K < 4$.

b) Assuming that the closed-loop is stable ($-2 < K < 4$), we can apply the Final Value Theorem

$$e(\infty) = \lim_{s \to 0} s\,(1 - G_c(s))\,R(s) = \lim_{s \to 0} s \frac{s^2 + 2s + 1}{s^2 + (2 + K)s + 1 - 0.25K} \frac{R}{s} = \frac{1}{1 - 0.25K} R$$

The output error is less than 10% of R if

$$\left| \frac{1}{1 - 0.25K} \right| \leq 0.1$$

or, equivalently,

$$|1 - 0.25K| > 10.$$

This inequality is satisfied by $K < -36 \vee K > 44$. This clearly violates the closed-loop stability assumption ($-2 < K < 4$). Therefore, there exists no K such that the error is less than 10%.

Problem 3 (6/30)

During your internship at Motormeme, a (fictitious) company designing electric motors, you are assigned the task of designing a proportional controller for a mass-driving motor. While the transfer function $G(s)$ is unknown, its Bode plot is available in Figure 2.

a) Design the gain $K > 0$ of the proportional controller such that all of the following requirements are satisfied

- The closed-loop bandwidth ω_B (approximated by the crossover frequency ω_c) is in the range $[8, 40]$ rad/s.
- The phase margin is at least $45°$.

(3p)

The system is affected by a measurement noise $n(t) = \sin(100t)$.

b) Within the range of values of K found at point (a), for what values of K does the closed-loop attenuate this disturbance by a factor larger than 5?

(3p)

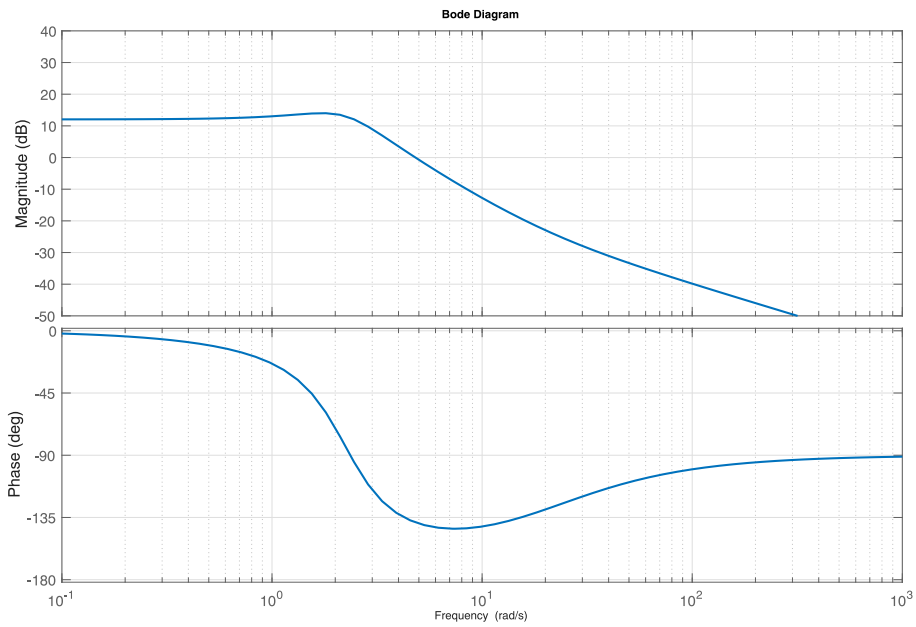

Figure 2: Bode plot of $G(s)$.

Proposed solution to problem 3

a) First, we note that $G_o(i\omega) = KG(i\omega)$. Since $K > 0$ is a real number,

- $|G_o(i\omega)| = \underbrace{20\log K}_{K_{\text{dB}}} + |G(i\omega)|$. That is, the magnitude of $G(i\omega)$ is shifted up/down by K_{dB} decibels.
- $\angle G_o(i\omega) = \underbrace{\angle K}_{0} + \angle G(i\omega) = \angle G(i\omega)$. That is, the phase of $G(i\omega)$ remains the same of Figure 2.

By inspecting Figure 2 we can notice that

- $|G(i8)| \approx -10\text{dB}$
- $|G(i40)| \approx -30\text{dB}$
- $\angle G(i40) > -135°$ for approximately $\omega > 14$ rad/s.

Therefore

- $\omega_c \geq 8$ rad/s if $K_{dB} > 10$ dB. This means $K > 10^{\frac{10\text{dB}}{20}} = 3.15$
- $\omega_c \leq 40$ rad/s if $K_{dB} < 30$ dB. This means $K < 10^{\frac{30\text{dB}}{20}} = 31.5$
- The phase margin $\varphi_m > 45°$ if $\omega_c > 14$ rad/s. Since $|G(i14) \approx -20$ dB, this means $K_{\text{dB}} \geq 20$ dB, which corresponds to $K \geq 10$.

Therefore, any $K \in [10, 31.5]$ satisfies the requirements.

b) The attenuation/amplification factor of the measurement noise is determined by the modulus, at $\omega = 100$ rad/s, of the complementary sensitivity function

$$|T(i100)| = \frac{|KG(i100)|}{|1 + KG(i100)|}$$

In the range $K \in [10, 31.5]$, $|KG(i100)| \ll 1$. This means that the denominator is dominated by 1 (and indeed we know that the complementary sensitivity function is well approximated by the open-loop transfer function for $\omega > \omega_c$). Therefore:

$$|T(i100)| = |K| \cdot |G(i100)| = 0.01K,$$

where $|G(i100)| = -40\text{dB} = 0.01$ has been derived from Figure 2. To have an attenuation factor larger than 5, we need

$$|T(i100)| = 0.01K < \frac{1}{5} \implies K < 20.$$

Problem 4 (6/30)

Consider a system which is on the state space form

$$\dot{x} = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} x.$$

a) What is the transfer function of the system from u to y?

(2p)

b) Design a linear state feedback of the form $u = \ell_1 x_1 + \ell_2 x_2$ that places the closed-loop poles at -3 and -2.

(2p)

c) Design a state observer that has both its eigenvalues in -5. Comment on the result.

(2p)

Proposed solution to problem 4

a) The transfer function is given by

$$G(s) = C(sI - A)^{-1}B. \tag{1}$$

Inserting the given A, B,and C yields

$$G(s) = \frac{s+1}{s^2+3s+2} = \frac{1}{s+2} \tag{2}$$

b) With $L = \begin{bmatrix} \ell_1 & \ell_2 \end{bmatrix}$ we get that $u = Lx$. and that the poles of the closed-loop system is the eigenvalues to $A + BL$. Using the normal procedure, we get $\ell_1 = -2$ and $\ell_2 = 4$. The characteristic polynomial is

$$\begin{aligned} \det\left(\lambda I - (A + BL)\right) &= \det \begin{bmatrix} \lambda + 3 - \ell_1 & 2 - \ell_2 \\ -1 & \lambda \end{bmatrix} \\ &= \lambda(\lambda + 3 - \ell_1) + (2 - \ell_2) \\ &= \lambda^2 + (3 - \ell_1)\lambda + (2 - \ell_2) \end{aligned} \tag{3}$$

If the poles are in -3 and -2 we get the characteristic polynomial

$$(\lambda + 3)(\lambda + 2) = \lambda^2 + 5\lambda + 6. \tag{4}$$

Matching coefficients of (3) and (4) give the equation system

$$\begin{aligned} 3 - \ell_1 &= 5 \\ 2 - \ell_2 &= 6, \end{aligned} \tag{5}$$

which has the solution $\ell_1 = -2$ and $\ell_2 = -4$.

c) Investigating the observability matrix $\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix}$, the system can be seen to be not observable since $\mathcal{O}$ is not full rank. Hence, we are not able to place the poles of the observer arbitrary. For example, it is impossible to place the poles in -5.

Problem 5 (6/30)

Over coffee your friend overshares that they are struggling with controlling a system of the form

$$\ddot{y}(t) = -\dot{y}(t) + u(t).$$

a) What is the transfer function from u to y of this system?

(1p)

b) You conclude that the desired bandwidth/cross-over frequency is ≈ 0.8 rad/s. What is a suitable sampling frequency for this system? Motivate your answer.

(1p)

c) Since you have taken a basic control course, you suggest that your friend should use a PD-controller with the transfer function

$$F(s) = K_d s + K_p.$$

Your friend replies with "a transfer function what?!" and wants you to provide a control law that can be implemented. Use Tustin's method (the bilinear transform with $\alpha = 0.5$) to discretize the controller. Use a sample time of 0.2 seconds. Answer with a control law in the form

$$u_k = ae_k + be_{k-1} + cu_{k-1}$$

so your friend can implement it. Note that a,b,and c, might depend on the free parameters K_d and K_p

(4p)

Proposed solution to problem 5

a) Laplace transform of the equation gives

$$s^2Y(s) = -sY(s) + U(s) \Leftrightarrow Y(s) = \frac{1}{s(s+1)}U(s). \tag{6}$$

Hence, the transfer function from u to y is $G(s) = \frac{1}{s(s+1)}$.

b) We want the sampling frequency to be higher than the bandwidth of the system. The bandwidth is

$$\approx 0.8 \text{ rad/s} = 0.8 * /(2\pi) \text{ Hz} \approx 0.13 \text{ Hz} \tag{7}$$

A good rule of thumb is to pick the sampling frequency about 20 to 50 times faster than the bandwidth. Hence, a suitable sampling frequency is $0.13 \cdot 35 \approx 4.5$ Hz.

c) With Tustin's method we replace

$$s = \frac{2}{T}\frac{1-q^{-1}}{1+q^{-1}}. \tag{8}$$

Since we are given $T = 0.2$ seconds, we get

$$s = 10\frac{1-q^{-1}}{1+q^{-1}}. \tag{9}$$

Inserting this in $F(s) = K_d s + K_p$ gives

$$F_d(q) = K_d 10\frac{1-q^{-1}}{1+q^{-1}} + K_p = \frac{K_p + 10K_d + (K_p - 10K_d)q^{-1}}{1+q^{-1}}. \tag{10}$$

The corresponding difference equation is

$$u_k + u_{k-1} = (K_p + 10K_d)e_k + (K_p - 10K_d)e_{k-1} \tag{11}$$

which gives the control law

$$u_k = (K_p + 10K_d)e_k + (K_p - 10K_d)e_{k-1} - u_{k-1}, \tag{12}$$

corresponding to $a = K_p + 10K_d, \quad b = K_p - 10K_d, \quad c = -1$.